M101 · Unit 4

Functions, limits and continuity

Recall: retrieve readily · Understand: explain the conditions · Consult: look up while working

Use	Topic	Essential statement or condition
Recall	**Function and domain**	One output for each allowed input; range: all attained outputs. Check denominators $\neq 0$, real square-root arguments ≥ 0, logarithm arguments > 0.
Understand	**Composition**	$(f \circ g)(x) = f(g(x))$. Require $x \in D_g$ and $g(x) \in D_f$. Order matters.
Understand	**Finite limit**	$f(x) \to L$ as $x \to a$: nearby nonzero offsets make outputs as close to L as required. The value at a need not exist or equal L.
Recall	**One-sided test**	At an interior point, a finite two-sided limit exists exactly when both one-sided limits exist and agree. At an interval endpoint, use the approach within the domain.
Understand	**Continuity**	At an interior point: $f(a)$ exists, the finite two-sided limit exists, and $\lim_{x \to a} f(x) = f(a)$. At included endpoints use the inward one-sided limit. Check each join separately.
Recall	**Finite-limit arithmetic**	If $f(x) \to A$ and $g(x) \to B$: $f \pm g \to A \pm B$, $cf \to cA$, $fg \to AB$, $\frac{f}{g} \to \frac{A}{B}$ $(B \neq 0)$. All component limits use the same approach and a common nearby domain.
Understand	**Continuous outer function**	If $g(x) \to L$ and f is continuous at L: $f(g(x)) \to f(L)$. Nearby inner outputs must lie in the outer domain.
Recall	**Elementary families**	Polynomials: continuous everywhere. Rational functions: denominator nonzero. $\sqrt{x}$: $x \geq 0$; $\ln x$: $x > 0$. Absolute value, sine, cosine and exponential: continuous on their real domains.

Teaching: CC BY-NC-SA 4.0. Code: MIT. LibraUni names and branding excluded.

Methods and numerical checks

Use	Topic	Essential statement or condition
Understand	**Equality near the target**	Equal rules on a deleted neighbourhood have the same limit. Cancellation preserves exclusions; a continuous extension is a separate definition.
Consult	**Algebraic methods**	Try valid substitution first. For $0 / 0$: factor, expand, combine fractions or multiply by a nonzero conjugate. $$(A - B)(A + B) = A^2 - B^2 .$$ $0 / 0$ is an indeterminate form, not an answer.
Recall	**Absolute value**	$$\lvert u \rvert = \begin{cases} u, & u \geq 0, \\ -u, & u < 0 . \end{cases}$$ Split at a sign change before cancelling.
Understand	**Failure and numerical evidence**	Unequal one-sided limits, unbounded growth or persistent oscillation can prevent a finite limit. A finite table or plot is evidence, not proof. Rounded subtraction of nearly equal numbers can lose accuracy.
Consult	**Stable example and domain**	$$\frac{\sqrt{9 + h} - 3}{h} = \frac{1}{\sqrt{9 + h} + 3} \to \frac{1}{6} \quad (h \to 0) .$$ Equality on the original domain: $h \geq -9$, $h \neq 0$. A finite-offset difference from $1 / 6$ is not necessarily rounding error.

Consult · Python quick lookup

```
from math import sqrt

def quotient(h):
    if h < -9 or h == 0:
        return None
    return 1 / (sqrt(9 + h) + 3)

for h in [0.01, -0.01, 1e-8, -1e-8]:
    print(h, quotient(h))
```

`None`: excluded input. Keep domain guards after rearranging.
Compare both signs of the offset; use algebra to justify the limit.

Notation: a: target input; L, A, B: finite limits; h: nonzero offset; D_f: domain of f.
Examples dimensionless. Retain physical units; logarithm/exponential arguments must be dimensionless.

Teaching: CC BY-NC-SA 4.0. Code: MIT. LibraUni names and branding excluded.